currents during the monsoons. Ferries and bridges made of boats usually sufficed, but the railways needed uninterrupted service and safety was essential to its success. There were many mistakes with tragic outcomes, before early engineers understood that European bridge-building techniques could not be replicated in the Indian subcontinent.

Battling the elements

Builders realized that they had to beat the rainy season or they could lose an entire year's work. This meant that they had eight to nine months to sink wells, build piers, and secure the foundations. This scramble required expensive equipment and efficient working systems. More than this, however, the engineers learnt how deep the foundations had to be for the bridges to withstand the onslaught caused by torrential rains and swollen rivers.

One of the biggest failures of this period includes the Narmada River bridge built by the Bombay, Baroda and Central India (BB&CI) Railway. Floods damaged it twice during construction. After it opened in 1861, it was damaged thrice over and lost three piers in 1868. So, too, did the bridges over the Jamuna, Sutlej, and Beas in the north. The last ended in tragedy after a passenger train fell into the raging river below.

Raw materials

The nature of Indian rivers necessitated a combination of iron and steel, unlike the brick or stone arch bridges on land whose piers were comparatively more vulnerable.

India’s first major railway bridge, the Tannah Viaduct, linked Thana and Kalyan, near Bombay, and was completed in 1854. Inspired by the ancient Romans, its style featured curved abutments and arch-shaped supports. The design allowed for load forces to be pushed along the curves of the arches and be carried by the abutments in a horizontal thrust, lessening the pressure on the overall structure. A similar arch-bridge was the Dapoorie Viaduct built in 1854. Joining the Bombay Island to the mainland, it comprised 22 stone arches, each spanning around 10 m. Today, it has been in use for more than 150 years. Much